Supplementary Material for: Resolving Sequence-Structure Misalignment in Antibody Design via Consistency-Aware Flow Matching

Appendix A
Model Implementation and Dataset Preparation

A. Network Architecture Overview

To ensure full reproducibility and address the details of our model design, we provide a comprehensive overview of the CCFlow architecture. Our base architecture is built upon a **Geometric Attention Network**, which strictly preserves $SE(3)$ equivariance while co-designing sequence and structure.

The forward generative process does not rely on recurrent discrete decoding but operates via a continuous-time Flow Matching ODE solver. Specifically, the network processes the noisy sequence logits (s_t) and 3D coordinates (R_t, p_t) at timestep t through the following stages:

1) **Feature Initialization:** The input states are first encoded into 1D node features and 2D pair features. Crucially, to prevent data leakage during training, context masking is rigorously applied to separate the generated regions from the conditioning anchor regions.
2) **Geometric Attention Encoder:** The core spatial reasoning is performed by a stack of `GABlock` layers. Each block updates the node representations by aggregating three types of attention:
   - 1D Node-wise sequence attention.
   - 2D Pair-wise distance/topology attention.
   - 3D Spatial attention, which is computed by projecting queries and keys into global 3D coordinates and deriving geometric point distances.
3) **Vector Field Prediction:** The updated $SE(3)$-invariant features are fed into independent Multi-Layer Perceptron (MLP) projection heads to concurrently predict the velocity vector fields for:
   - Sequence logits (vel_s $\in \mathbb{R}^{L\times 20}$)
   - $C\alpha$ coordinate translations (vel_pos $\in \mathbb{R}^{L\times 3}$)
   - Frame orientations updated via Grassmann products (vel_qua $\in \mathbb{R}^{L\times 4}$)
   - Full backbone atom positioning (vel_mainchain $\in \mathbb{R}^{L\times 4\times 3}$)

B. Parameterization Details of Key Modules

Importantly, all encoders in CCFlow are jointly learned from scratch during the flow matching training, rather than relying on frozen pre-trained language or structure models.

- **Residue Encoder (h_{seq} and h_{struct}):** Both sequence and structural features are jointly parameterized within the `ResidueEmbedding` module. The sequence representation (h_{seq}) is initialized via a learned `nn.Embedding` layer over the 20 amino acid types. The structural representation (h_{struct}) is parameterized by explicitly mapping global 3D coordinates into local frames, augmented with backbone dihedral angles encoded via continuous `AngularEncoding`. These modal features, along with fragment type embeddings, are concatenated and fused through a 4-layer MLP (with ReLU activations) to produce the unified 1D node feature.
- **Pair Encoder (h_{pair}):** Parameterized by the `PairEmbedding` module, the 2D pair feature captures the topological graph. It fuses: (1) learned embeddings of amino acid pairs, (2) relative sequential positions, (3) pairwise spatial distances encoded via Gaussian Radial Basis Functions (RBFs), and (4) pairwise dihedral angles. These are concatenated and refined through a 3-layer MLP.
- **Cross-Modal Alignment Projections (ϕ, ψ):** To compute the latent alignment loss (L_{Align}), the sequence node features are projected into a shared semantic space. Specifically, ϕ is parameterized as a linear projection layer that maps the sequence embedding dimension to match the residue feature dimension, while ψ acts as an identity mapping since the target structure features are already in the correct dimensionality. To capture local sequence-structure motifs, we additionally employ a local contextual encoder parameterized as a 1D Convolutional layer with a kernel size of 3 and no bias, operating directly on the aligned latent space.
- **Geometric Implication Projector (G):** In response to questions regarding the nature of the geometric projector G, we clarify that it is implemented as a jointly learned auxiliary regression head, rather than a fixed mathematical mapping. Specifically, G consists of a 1D Convolutional context layer (mapping the 20-dimensional amino acid probabilities to 64 channels) followed by a 2-layer MLP (Hidden dimension 128, ReLU activation). It maps the sequence logits to predict four core 1D geometric properties of the $C\alpha$ backbone: adjacent distances, planar angles, and the sine/cosine of torsion angles. It is supervised using a Huber loss ($\delta = 1.0$) against the actual geometric targets extracted from the dynamically

generated 3D coordinates.

- **Contact Alignment Module:** For the contact alignment module, the pairwise features are projected via a 2-layer MLP (hidden dimension 64) into a scalar logit. The target contact map is defined using a strict 8 Å $C\alpha$ distance threshold, which is the standard definition of residue-residue contact in structural biology protocols.

C. Dataset Curation and Homology Removal

To ensure a rigorous evaluation and prevent data leakage, we carefully curated the training samples derived from the Structural Antibody Database (SAbDab). A strict homology removal protocol was implemented using the MMseqs2 easy-cluster module to guarantee that the model's performance on the test set reflects true generalization rather than the memorization of homologous training samples. Specifically, we applied stringent clustering hyperparameters, including a minimum sequence identity threshold of 50%, a sequence coverage threshold of 80%, and a coverage mode requiring the alignment to cover 80% of the shortest sequence.

To construct our training, validation, and test splits without information leakage, we executed a cluster-aware exclusion strategy specifically targeting the RAbD test set. We first mapped all antibodies from the test set to their corresponding MMseqs2 clusters. Subsequently, any cluster containing at least one RAbD test sample was entirely excluded from the candidate pool for the training and validation sets. The remaining pristine clusters were then partitioned to form the final training and validation datasets. Through this rigorous cluster-based exclusion mechanism, we guarantee that no antibody in our training set shares a sequence identity greater than 50% with any target in the RAbD test set, confirming that our proposed framework is evaluated under a realistic and highly challenging zero-shot design scenario. This rigorous data curation protocol aligns with established practices in recent antibody design literature, such as dyAb and dyMEAN.

Appendix B
Comparative Analysis of Model Complexity and Efficiency

To provide a comprehensive understanding of our proposed CCFlow relative to existing antibody design baselines, we conduct a detailed comparative analysis focusing on parameter efficiency, sampling complexity, and computational resource requirements.

A. Model Parameterization and Scale

The statistical results of total parameters and model sizes are summarized in Table I.

Our model maintains a parameter count (~4.3M) comparable to DiffAb and FlowDesign. This ensures that the performance gains are derived from the algorithmic consistency and anchor-aware curriculum learning rather than simple parameter scaling. AbX utilizes ESM-2 (3B) as its backbone, resulting in a parameter count approximately 660× larger than our approach. Such a massive scale poses significant challenges for deployment on consumer-grade GPUs and real-time design tasks.

TABLE I
Comparison of model parameters and storage sizes across different antibody design frameworks.

Model	Architecture	Params (M)	Size (MB)
MEAN	Equivariant GNN	2.00	7.66
dyMEAN	Dynamic GNN	2.01	7.66
DiffAb	Diffusion	4.00	15.24
FlowDesign	Rectified Flow	4.19	16.00
CCFlow (Ours)	**Consistent Flow**	**4.30**	**16.39**
AbX	ESM-2 + Diffusion	2850.95	10875.51

B. Inference Efficiency and NFE

The Number of Function Evaluations (NFE) is a critical metric for evaluating the throughput of antibody design in practical wet-lab scenarios.

- **One-step Models (NFE=1):** MEAN and dyMEAN employ a direct regression approach. While extremely fast, they often struggle to capture the complex structural distribution of CDR loops compared to iterative generative methods.
- **Iterative Generative Models (NFE=100):** DiffAb, FlowDesign, and our CCFlow typically use 100 steps for high-fidelity sampling.
- **Computational Bottleneck:** Although AbX also uses 100 steps, each step involves an inference pass through the massive ESM-2 transformer, leading to the highest latency and lowest throughput among all tested methods.

C. Computational Resource Requirements

We evaluate the peak GPU memory (VRAM) usage and relative latency during the inference phase with standard batch sizes as shown in Table II.

TABLE II
Inference efficiency and peak resource consumption.

Model	Batch Size	NFE	Peak VRAM	Rel. Latency
MEAN / dyMEAN	16	1	~4–6 GB	1.0×
DiffAb	16	100	~16 GB	~110×
FlowDesign	8	100	~12–16 GB	~100×
CCFlow (Ours)	**6**	**100**	**~10–14 GB**	**~95×**
AbX	1	100	$>$ 16 GB	$>$ 500×

Despite the iterative nature of Rectified Flow, our implementation demonstrates slightly better VRAM efficiency and inference speed compared to the standard Diffusion-based DiffAb, benefiting from more efficient pair-feature processing and the deterministic nature of the flow ODE.

D. Ablation Study on Sampling Trajectories

To deeply understand the generative dynamics of our framework, we conducted a comprehensive trajectory analysis during the inference phase. Specifically, the data presented in Table III is derived from the generative evaluation of the 5NUZ antibody sample from RAbD set. It records the intermediate states of the reverse Ordinary Differential Equation (ODE) sampling process, tracking the continuous evolution from the

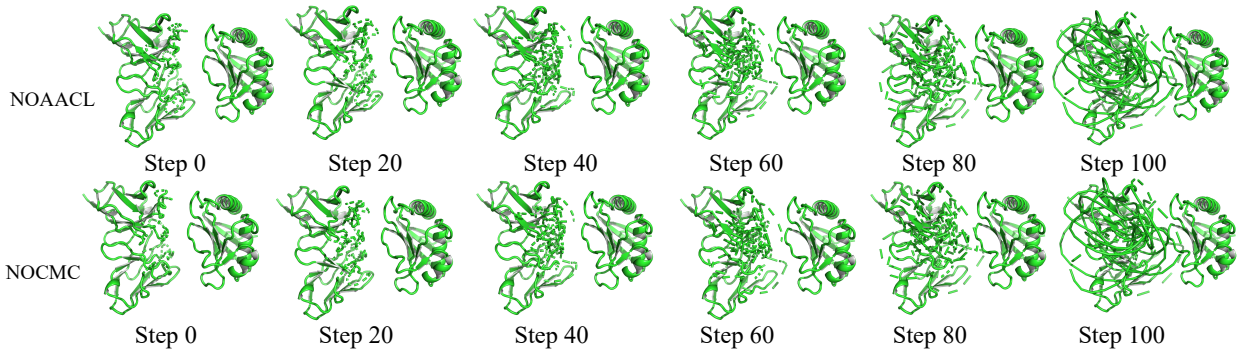

Fig. 1. Structural evolution trajectories of the 5nuz complex generated by the two ablated models.

initial Gaussian noise distribution ($t = 100$) to the final generated antibody structure and sequence ($t = 0$).

By comparing the CCFlow model against two ablation variants, namely no CMC (removing Cross-Modal Consistency) and no AACL (removing Anchor-Aware Curriculum Learning), we aim to explicitly isolate and validate the specific contributions of these proposed modules to the overall structural convergence and sequence recovery on this representative target.

E. Ablation Study on Generation Dynamics

To further evaluate the contributions of individual components, we conducted an identical trajectory analysis on two ablated versions of the CCFLOW model: one removing the cross-modal consistency constraint and the other omitting the anchor-aware curriculum learning module. As shown in Figure 1, intermediate structures were similarly extracted at time steps 0, 20, 40, 60, 80, and 100. By comparing these generation dynamics against the full model, we can visually assess how the absence of these key modules impacts the folding stability, assembly efficiency, and final structural precision of the 5NUZ.

TABLE III

DETAILED ABLATION STUDY ON THE SAMPLING TRAJECTORY ACROSS ALL SIX CDRS. WE REPORT RMSD (↓, IN Å) AND AAR (↑, IN %) EVALUATED FROM THE INITIAL NOISE STATE ($t = 100$) TO THE FINAL GENERATED SAMPLE ($t = 0$).

Step (t)	**CDR**	**No AACL**		**No CMC**		**CCFlow**	
		RMSD ↓	**AAR ↑**	**RMSD ↓**	**AAR ↑**	**RMSD ↓**	**AAR ↑**
100 (Noise)	H1	14.2585	28.57%	16.4599	42.86%	16.0266	28.57%
	H2	13.1406	16.67%	16.9405	50.00%	13.7101	33.33%
	H3	14.8768	12.50%	14.3519	18.18%	14.5072	18.75%
	L1	12.4561	23.08%	19.7767	20.00%	12.4160	30.77%
	L2	11.9239	28.57%	15.5262	28.57%	13.1478	28.57%
	L3	15.3261	27.27%	14.6040	33.33%	14.2167	27.27%
80	H1	11.5600	28.57%	13.3641	42.86%	12.7290	28.57%
	H2	10.6005	16.67%	13.6569	50.00%	10.9068	33.33%
	H3	11.9257	12.50%	11.5907	27.27%	11.7339	18.75%
	L1	10.0358	30.77%	15.8594	20.00%	9.9429	30.77%
	L2	9.6093	28.57%	12.3719	28.57%	10.4801	28.57%
	L3	12.3222	27.27%	11.7036	33.33%	11.4257	27.27%
60	H1	8.7468	28.57%	10.1448	42.86%	9.4388	28.57%
	H2	8.0301	50.00%	10.2246	50.00%	8.0742	33.33%
	H3	8.9946	18.75%	8.6335	18.18%	9.0913	25.00%
	L1	7.6114	38.46%	11.8502	20.00%	7.4726	38.46%
	L2	7.2392	42.86%	9.0826	28.57%	7.7724	28.57%
	L3	9.3306	36.36%	8.6839	33.33%	8.6657	36.36%
40	H1	5.9923	42.86%	6.9414	71.43%	6.2499	57.14%
	H2	5.6424	50.00%	6.7902	50.00%	5.4335	33.33%
	H3	6.1532	31.25%	5.9362	27.27%	6.6818	25.00%
	L1	5.2226	46.15%	7.9332	26.67%	5.1808	53.85%
	L2	4.7504	42.86%	5.8991	42.86%	5.2208	42.86%
	L3	6.5077	45.45%	5.7813	55.56%	6.0698	36.36%
20	H1	3.5047	57.14%	4.2906	100.00%	3.5807	85.71%
	H2	3.9723	50.00%	3.9475	50.00%	3.6430	50.00%
	H3	3.7184	37.50%	4.2182	27.27%	4.5592	31.25%
	L1	3.4035	76.92%	4.6443	46.67%	3.4060	53.85%
	L2	2.7131	42.86%	3.3126	57.14%	3.1062	57.14%
	L3	4.0343	45.45%	3.4108	66.67%	3.9221	45.45%
0 (Final)	H1	3.2729	71.43%	3.4882	85.71%	2.1802	85.71%
	H2	3.9939	50.00%	2.7674	50.00%	3.4360	50.00%
	H3	3.2456	31.25%	5.1571	36.36%	3.9212	43.75%
	L1	3.6014	61.54%	3.5060	46.67%	2.9642	84.62 %
	L2	2.2318	42.86%	2.7885	57.14%	2.8734	57.14%
	L3	3.4552	54.55%	3.0938	66.67%	3.1168	45.45%